Analyzing the Impact of Economic Factors on Housing Prices in the United States.

Economic Drivers of U.S. Housing Prices

Stanley Occean & Jonah Liautaud
Pace University – CS668

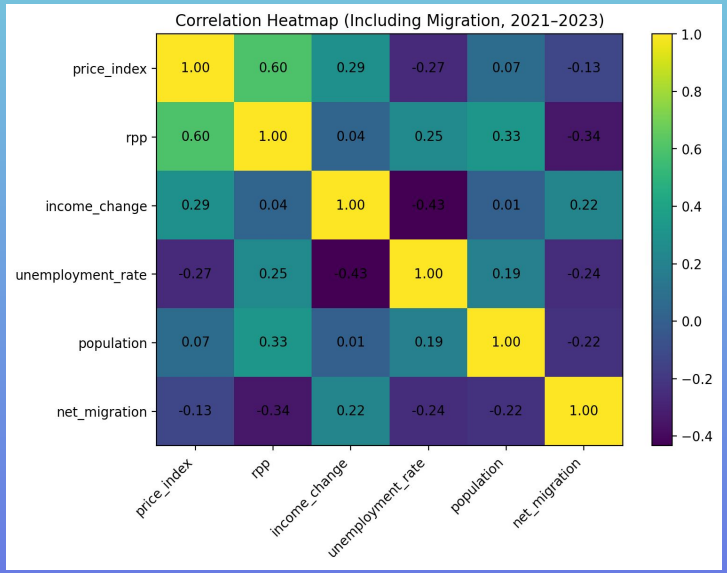
Problem Statement

- Housing prices affect affordability, migration, and financial stability.
- Economic conditions vary widely across U.S. states.
- States differ in income growth, unemployment, population change, migration, and cost of living.
- Research question: How are these factors associated with housing prices from 2020 to 2024?
- Main goal: explain housing prices using a multivariable approach.

New EDA

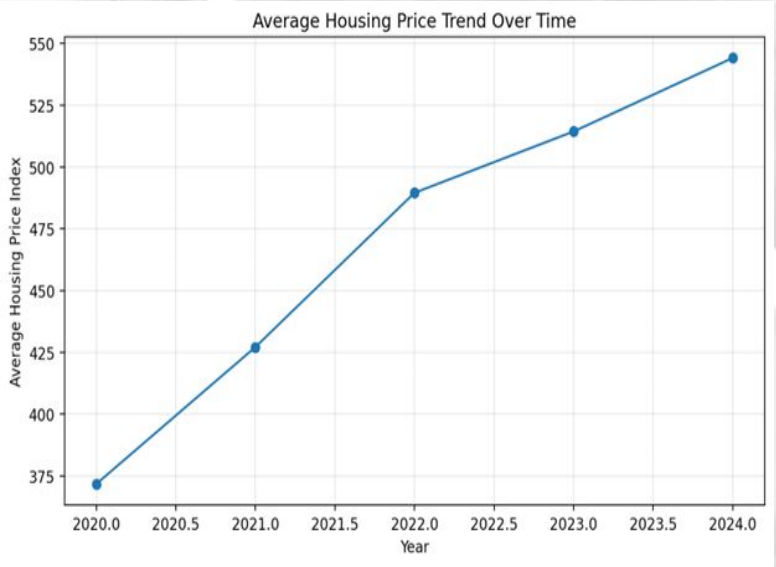
- Housing prices increased (2020 → 2024)
- RPP strongest positive correlation (~0.60)
- Unemployment negative relationship
- Income positive but weaker
- Migration weakest

Correlation Heatmap



Correlation heatmap of housing price index and key economic variables, including RPP, income change, unemployment rate, population, and net migration (2021–2023). The results show that RPP has the strongest positive relationship with housing prices, while the unemployment rate exhibits a negative correlation.

AVerage Housing Price



Trend of average housing price index from 2020 to 2024. The steady increase reflects strong growth in housing prices across U.S. states during this period.

Propose Solution

Built a 50-state panel dataset from 2020 to 2024.

Used multiple linear regression as the main model.

Used Random Forest Regression as a comparison model.

Combined data from BLS, Census, BEA, FHFA, and IRS.

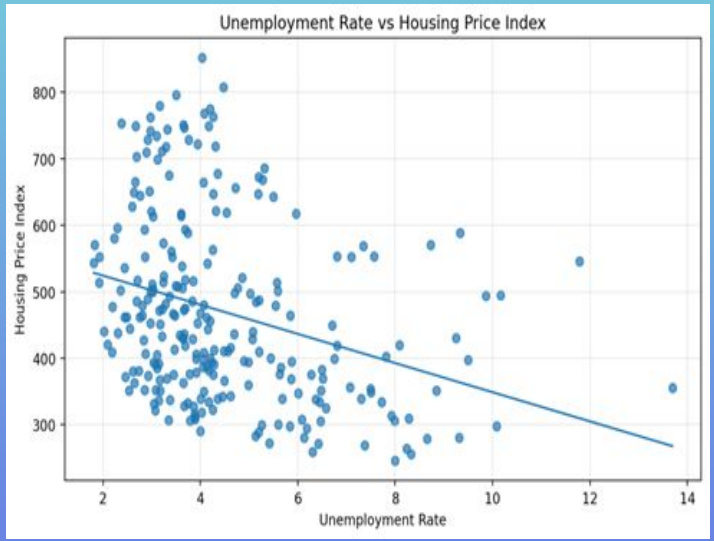
Migration was only available for the reduced 2021 to 2023 subset.

Housing Prices

Result and Analysis 1

- RPP was the strongest associated factor.
- Correlation with housing prices was about 0.56 to 0.60.
- Higher-cost states generally had higher housing prices.
- Unemployment was negatively associated with housing prices.
- Income growth had a positive relationship.
- Population and migration were weaker predictors.

Unemployment Rate VS Housing Price Index



Scatter plot of unemployment rate versus housing price index. The downward trend line indicates a negative relationship, suggesting that higher unemployment rates are associated with lower housing prices.

Income Change VS Housing Price Index

Scatter plot of income changes versus housing price index. The upward trend line shows a positive relationship, indicating that higher housing prices are associated with higher income, even if the effect is moderate

Table X. Model Performance Comparison

Model	Dataset	RMSE	MAE	R^2
Multiple Linear Regression	2020–2024 (Baseline)	108.06	89.50	0.40
Random Forest Regression	2020–2024 (Baseline)	98.61	78.86	0.50
Multiple Linear Regression	2021–2023 (Migration Subset)	95.22	73.43	0.55
Random Forest Regression	2021–2023 (Migration Subset)	104.96	77.99	0.46

- Random Forest performed better on the full dataset.
- It had lower RMSE and MAE and higher R-squared.
- Multiple linear regression performed better on the smaller migration subset.
- Model performance depended on sample size and structure.
- No single model was best in every case.

Result And Analysis 2

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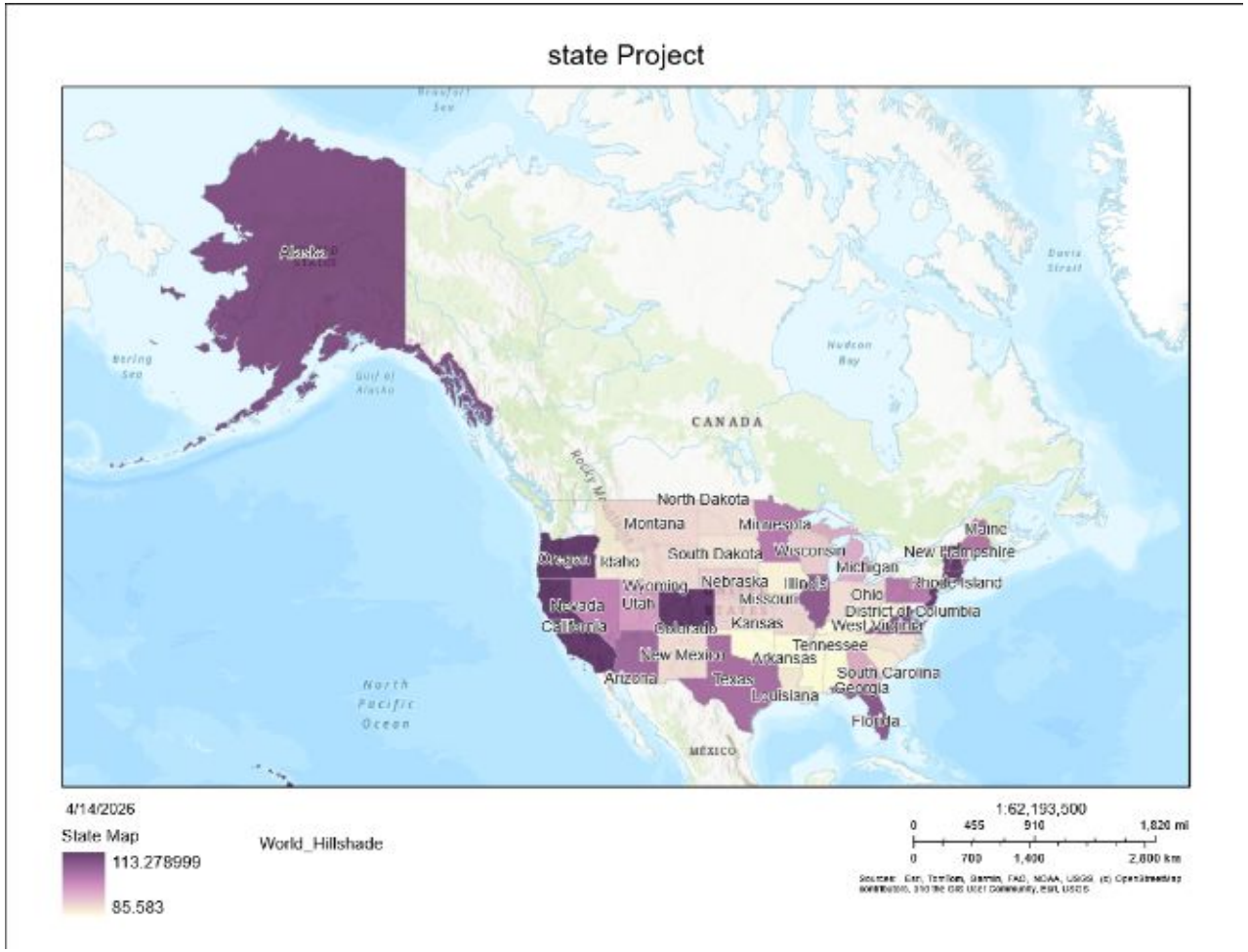
Choropleth Map

Choropleth map of housing price index across U.S. states. Darker shades indicate higher housing price levels, highlighting regional disparities, with higher concentrations in coastal and economically strong states.



Conclusion And future Work

- *Housing prices driven by multiple factors*
- *RPP is the strongest driver*
- *Income and unemployment also important*
- *Migration limited by data.*
- *Add interest rates and housing supply.*
- *Use city-level data*
- *Improve ML models*

References

- U.S. Census Bureau.
- Bureau of Labor Statistics.
- Federal Housing Finance Agency.
- Bureau of Economic Analysis.
- IRS migration data.